

Luis Mastrangelo

Senior Software Engineer

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Professional Summary

I am Senior Software Engineer with professional and academic experience in Switzerland, Argentina, France and USA. In the last years I have been working in the Web3 space. I worked with Ethereum, NEAR, Avalanche, EOS/WAX and Everscale (TON) using mainly Solidity, Rust and TypeScript/JavaScript. I have worked for more than 15 years in the private industry in different roles, as Software Engineer, Network Administrator and Freelance Developer. I delivered many software projects for a wide variety of clients across different sectors. Moreover, I have a Ph.D. in Informatics at the University of Lugano, Switzerland, where I was able to perform outstanding research in programming languages, runtimes and virtual machines.

Work Experience

- Nov 2023–Mar 2026 **Senior Software Engineer, Hashgraph (ex. Swirlds Labs), Remote**
Part of the Smart Contracts team, building tools to support EVM development in Hedera networks. While working here I
- created the Hedera Forking project, which allows developers to fork test their contracts against Hedera System Contracts,
 - contributed to setup a custom Sourcify instance, which allowed developers to verify their Smart Contracts,
 - refactored Hashgraph's JSON-RPC Relay so it can be more easily extensible,
 - integrated the Ethereum Execution Client test framework into Hedera's EVM,
 - debugged production issues while on-call using Grafana and GCP, and
 - added support to onboard new clients to Hashgraph's TheGraph custom instance.
- Apr 2023–May 2023 **Senior Software Engineer, ChainSafe, Remote**
I worked in Lodestar, an Ethereum Consensus Client written in TypeScript. I helped to fix some small bugs while working here.
- Jul 2022–Jan 2023 **Senior Blockchain Engineer, ConsenSys, Remote**
Part of the Truffle team, a development environment to build and test Solidity Smart Contracts. I focused on their VS Code extension for Truffle. While working here I
- added and tested new features to the extension using TypeScript, and
 - helped with the release workflow in GitHub Actions.
- May 2022–Jun 2022 **Lead Blockchain Engineer, Campground, Remote**
Smart Contract development with Rust and TypeScript for the NEAR blockchain. While working here I
- built a system in Rust to automatically generate public docs for our smart contracts,
 - tested and deployed the main smart contracts, and
 - helped with the integration of the Smart Contracts into our backend using TypeScript.

- Sep 2020–Apr 2022 **Senior Software Engineer**, *EPAM Systems*, Zurich, Switzerland
Ethereum & NEAR blockchains development with Solidity, Rust and TypeScript.
- Aug 2019–Jan 2020 **Technical Consultant**, *DB&L Services SA for Avaloq*, Lugano, Switzerland
IT infrastructure support for Monitoring Operations at Avaloq Sourcing (CH & LI) SA.
- Dec 2006–Mar 2011 **Project Leader**, *Recursiva S.R.L.*, Buenos Aires, Argentina
Software engineering and architect roles using Java and C# among others technologies.
- Mar–Jul 2006 **Network Administrator**, *Texpampa and HS Eventos*, Buenos Aires, Argentina
Technical service, software installation and maintenance of company network.
- Mar 2000–May 2004 **Software Developer**, *Fernando Hermo y Asoc.*, Buenos Aires, Argentina
ERP and CRM development using VB 6, SQL Server and Crystal Reports.

Education

- Apr 2013–June 2019 **Ph.D. Graduate, Computer Science**, *University of Lugano*, Switzerland
Programming languages research in Java, Scala and Haskell, runtime systems and virtual machines with C++, software verification, data analysis using R and Python.
- Apr 2007–Nov 2011 **Licentiate in Computer Science**, *University of Buenos Aires*, Argentina
This degree is considered by many institutions as being equivalent to an M.Sc. degree.
- Apr 2011–Oct 2011 **Internship**, *INRIA/University of Strasbourg*, France
Internship title: "A Virtual Machine for Automatic Program Parallelization".
- Apr 2003–Dec 2006 **Computer Analyst**, *University of Buenos Aires*, Argentina
This is a 3 years degree as part of the *Licentiate in Computer Science* programme.

Freelance & Open Source Projects

- Apr 2023–Mar 2025 **Software Engineer**, *SEVM Open Source project*, Remote
A Symbolic EVM bytecode interpreter, parser and decompiler written in TypeScript.
- Jan–May 2020 **Software Engineer**, *GTM (cont.) and GTM Dashboard Open Source*, Remote
Added support for submodules to Git Time Metric (Golang) PR and built a dashboard for GTM from scratch (Rust, TypeScript and Svelte).
- Feb 2014–Jan 2020 **Software Developer**, *JNIF*, Lugano, Switzerland
Java Native Instrumentation Framework written in C++.
- May–Sep 2018 **Software Developer**, *Essentia, Family Office*, Lugano, Switzerland
Desktop application written in C# and Scala.
- Apr 2016–Oct 2017 **Software Developer**, *HSC*, Remote
A Supercompiler prototype for Haskell.

Technical Skills

- Programming Languages Rust, JavaScript/TypeScript, Solidity/Yul, Java, Kotlin, Scala, C/C++, C#, Haskell, Golang, Python, R, SQL, PHP, HTML/CSS, UML
- Paradigms & Fields OOP, Functional Programming, Static & Dynamic Analysis, Data Science
- Databases PostgreSQL, MySQL, Oracle, Datalog, SQL Server, SQLite, Redis, Design
- Editors VSCode, IntelliJ/CLion/Rider, Eclipse, Visual Studio, Xcode, Vim/Emacs
- VCSs Git, GitHub, GitLab, BitBucket, Mercurial, Subversion, TortoiseSVN, CVS
- DevOps Docker, Podman, Kubernetes, Helm, Lima, GitHub Actions, GitLab CI/CD

Tools GitHub Copilot, Node.js, Apache/Nginx, Azure, GCP
Project Management Scrum, GitHub Projects, Agile, JIRA, Kanban Boards
Blockchain & Web3 Ethereum, NEAR, Avalanche, Smart Contracts, Foundry, Hardhat, TheGraph

Language Proficiency

English **Proficient User:** Level C2 under CEFR
Spanish **Native speaker**
Italian **Proficient User:** Level C1 under CEFR
French **Independent User:** Level B1 under CEFR

Teaching Experience

Apr 2013–Nov 2017 **Teaching Assistant**, *University of Lugano*, Switzerland. Multiple courses.
Apr 2006–Nov 2010 **Teaching Assistant**, *University of Buenos Aires*, Argentina. Multiple courses.

Publications

University of Lugano **Luis Mastrangelo**, Ph.D. Thesis, Advisors: Prof. M. Hauswirth and Prof. N. Nystrom. When and How Java Developers Give Up Static Type Safety. 2019
University of Lugano **Luis Mastrangelo**, M. Hauswirth and N. Nystrom. Casting about in the Dark: A Survey of Cast Usage in Java. 2019
University of Lugano **Luis Mastrangelo**, L. Ponzanelli, A. Mocci, M. Lanza, M. Hauswirth and N. Nystrom. Use at Your Own Risk: The Java Unsafe API in the Wild. 2015
University of Lugano **Luis Mastrangelo** and M. Hauswirth. JNIF: Java Native Instrumentation Framework. 2014
INRIA/University of Strasbourg A. Jimborean, **Luis Mastrangelo**, V. Loechner and P. Clauss. VMAD: an Advanced Dynamic Program Analysis & Instrumentation Framework. 2012
INRIA/University of Strasbourg A. Jimborean, P. Clauss, B. Pradelle, **Luis Mastrangelo** and V. Loechner. Adapting the Polyhedral Model for Efficient Speculative Parallelization. 2012